

Use-Case Flow — Release Escrow Payment

Use Case ID	UC-05
Use Case Name	Release Escrow Payment
Goal	To securely release the milestone payment held in escrow to the Freight Carrier's bank account after verified proof-of-delivery confirmation, ensuring both parties are financially protected.
Primary Actor	Payment Gateway
Secondary Actors	Shipper, Freight Carrier
Trigger	The Freight Carrier uploads a Proof-of-Delivery (POD) document for an awarded shipment.

Preconditions

1. The shipment contract has been awarded to a carrier (UC-03 completed).
2. The shipper has funded the escrow account with the agreed bid amount.
3. The freight carrier has physically delivered the cargo to the destination.
4. The freight carrier has uploaded a valid POD document (UC-04 completed).
5. Both shipper and carrier have verified and active bank account details on file.

Postconditions

1. The milestone payment has been transferred from the escrow account to the carrier's registered bank account.
2. The shipment status is updated to "Completed — Payment Released."
3. Both shipper and carrier have received a digital payment confirmation receipt via email and in-app notification.
4. The transaction is recorded in the system's financial audit log with a unique transaction reference ID.

Main Success Scenario

Step	Actor	Action
1	Freight Carrier	Uploads the digitally signed Proof-of-Delivery document via the shipment dashboard (includes recipient name, signature image, delivery timestamp, and optional photo evidence).
2	System	Validates the POD document

		format, checks for mandatory fields (recipient name, signature, timestamp), and verifies the digital signature integrity.
3	System	Updates the shipment status from "In Transit" to "Delivered — Pending Verification" and logs the POD submission timestamp.
4	System	Sends a notification (email + in-app) to the Shipper informing them that a POD has been uploaded and is awaiting their confirmation.
5	Shipper	Logs into the platform, navigates to the shipment dashboard, and reviews the uploaded POD document — inspecting the recipient signature, delivery timestamp, and photo evidence.
6	Shipper	Confirms the POD by clicking the "Approve Delivery" button, attesting that the cargo was received in satisfactory condition.
7	System	Records the shipper's approval with a timestamp and updates the shipment status to "Delivery Confirmed."
8	System	Initiates the escrow release workflow: calculates the net payable amount (bid amount minus platform commission), generates a payment instruction, and transmits it to the Payment Gateway via a secure API call.
9	Payment Gateway	Processes the payment instruction, debits the escrow holding account, and credits the Freight Carrier's registered bank account with the net payable amount.
10	Payment Gateway	Returns a transaction confirmation response to the system, including a unique transaction reference ID, transfer amount, and settlement timestamp.
11	System	Updates the shipment status to

		"Completed — Payment Released" and records the transaction details in the financial audit log.
12	System	Sends payment confirmation receipts to both the Shipper (showing amount deducted and platform commission) and the Freight Carrier (showing net amount received and transaction reference).

Alternate Flow — Auto-Approval After 72-Hour Window

Step	Actor	Action
A1 (branches from Step 5)	System	The shipper does not log in or take any action on the POD within the 72-hour review window.
A2	System	After 72 hours have elapsed since the POD notification, the system triggers the auto-approval mechanism and logs the event as "Auto-Approved — Shipper Inaction."
A3	System	Sends a final reminder notification to the shipper informing them that the POD has been auto-approved due to the expiry of the review window.
A4	System	Proceeds to Step 8 of the Main Success Scenario to initiate the escrow release workflow.

Exception Flow — Payment Gateway Failure

Step	Actor	Action
E1	Payment Gateway	Returns an error response indicating a transaction failure (e.g., insufficient escrow balance, carrier bank account invalid, gateway timeout).
E2	System	Logs the failure with the error code, transaction details, and timestamp in the financial audit log.
E3	System	Updates the shipment status to "Payment Failed — Retry Pending" and schedules an automatic retry after 30

		minutes.
E4	System	Sends an alert notification to the System Administrator with the failure details for manual investigation if needed.
E5	System	Retries the payment instruction (up to a maximum of 3 attempts). If successful, proceeds to Step 10 of the Main Success Scenario.
E6	System	If all retry attempts fail, updates the status to "Payment Failed — Manual Intervention Required" and notifies both the shipper and carrier of the delay, along with the expected resolution timeline.